Building Science

Lecture 1: Introduction

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Today’s outline

- Course content
- What to expect from this course
- What is ‘Building Science’?

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Course content:

1. General introduction to Building Science
2. Climate and Building orientation
3. Energy efficiency and Passive control strategies
4. Thermal performance – heat transfer
5. Thermal comfort
6. Thermal performance – Insulation and thermal bridges

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Course content (continued)

9. Natural lighting
10. Artificial lighting
11. Building envelope systems - Windows
12. Acoustics
13. Energy
14. Water
15. Summary of course

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Major themes we will learn about:

- General introduction to Building Science
- Climate impacts on Building orientation
- Energy efficiency and Passive control strategies
- Thermal performance – heat transfer principles
- Thermal performance – Insulation
- Thermal performance – Thermal bridges
- Thermal comfort
- Moisture control
- Air flow, ventilation and leakage
- Natural and artificial lighting
- Acoustics and noise control
- Building envelope systems – Windows
- Acoustics
- Energy management
- Water management

3x QUIZZES

1x EXAM

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What is it?

- The study of ***environmental factors*** affecting buildings
- Includes:
  - The <u>measurement</u> and <u>analysis</u> of these *factors*, and
  - The ways to <u>control</u> of these *factors.*

(e.g. Building *Services*)

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What is it?

- The science of optimising *building performance*, but also preventing *building failure*

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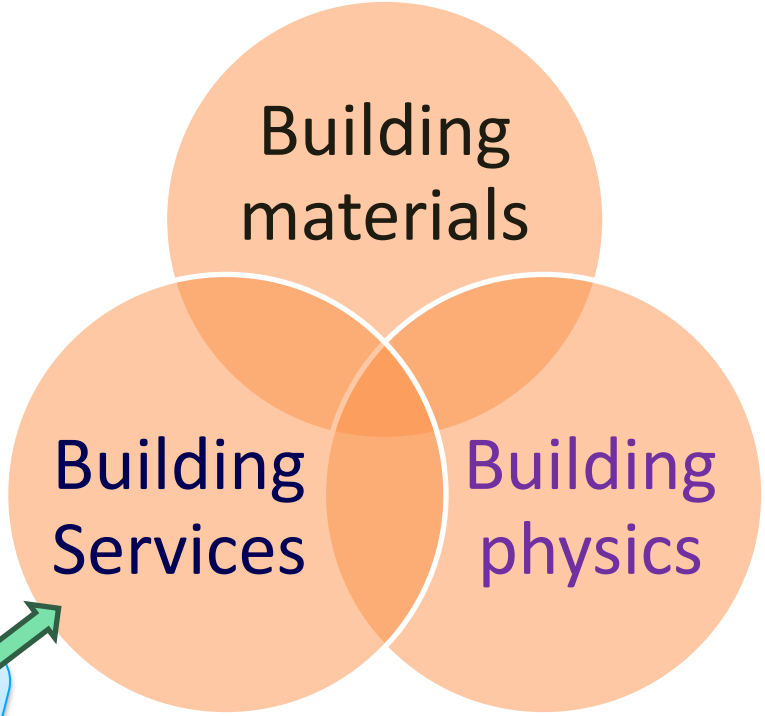
What is it?

- There is a lot to learn!
- *This course is just one part*



A separate course that you will learn about later

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Building Services	Building Physics	Building Materials
Ventilation	Thermal environment	Timber
Heating and cooling	Indoor air quality	Concrete
Electrical power	Acoustics	Brick masonry
Vertical transport	Visual environment	Stone
Fire protection	Energy/water usage	Metals
Water supply	Waste management	…
	…	

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Can you think of some examples?

Why is Building Science important?

- ✓Building systems can be simple
- ✓***But!*** Building systems interact in complex ways!
  - ✓The different parts of buildings are inter-related
  - ✓One small change can IMPACT many other parts in unexpected ways.

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Why is it important?

✓Buildings create *indoor environments* that are ideally comfortable FOR PEOPLE

What makes a space comfortable?

- Temperature
- Humidity
- Light
- Air quality

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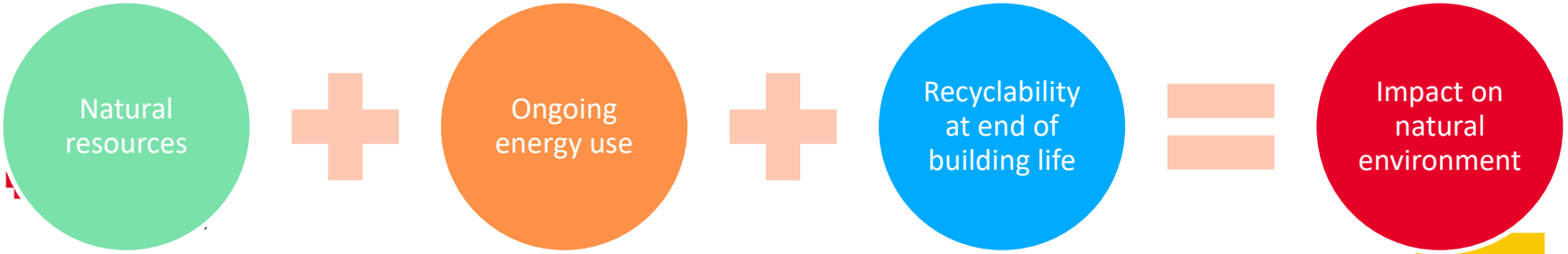
Why is it important?

- Also, buildings have a large impact on the *natural environment*.

- When creating buildings, we MUST minimise *environmental degradation* and *resource depletion*



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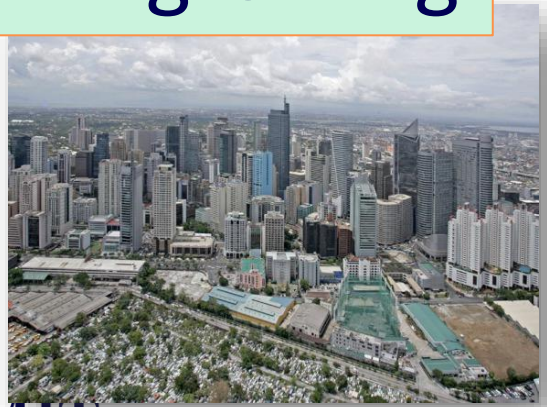
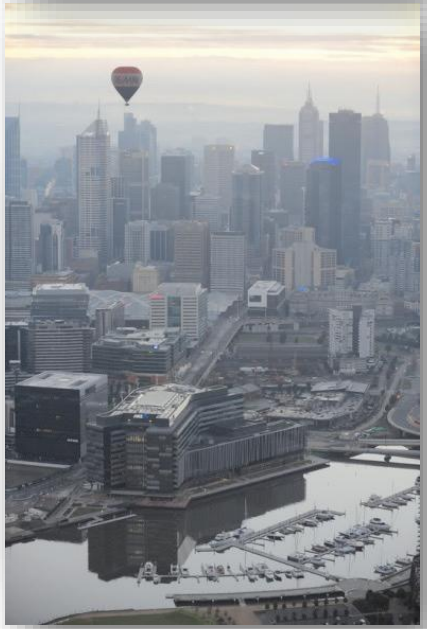
Why is it important?

Cities are growing





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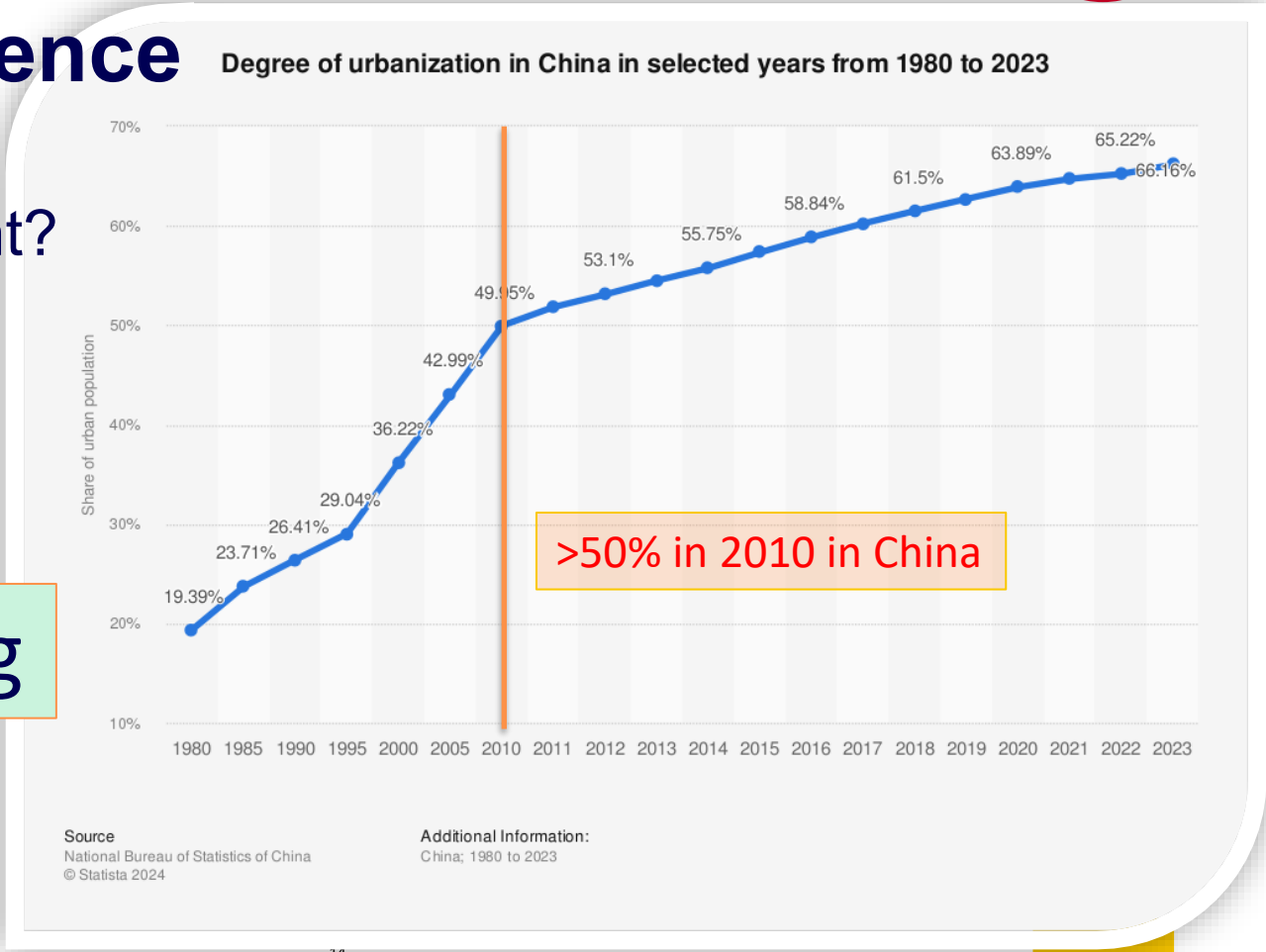

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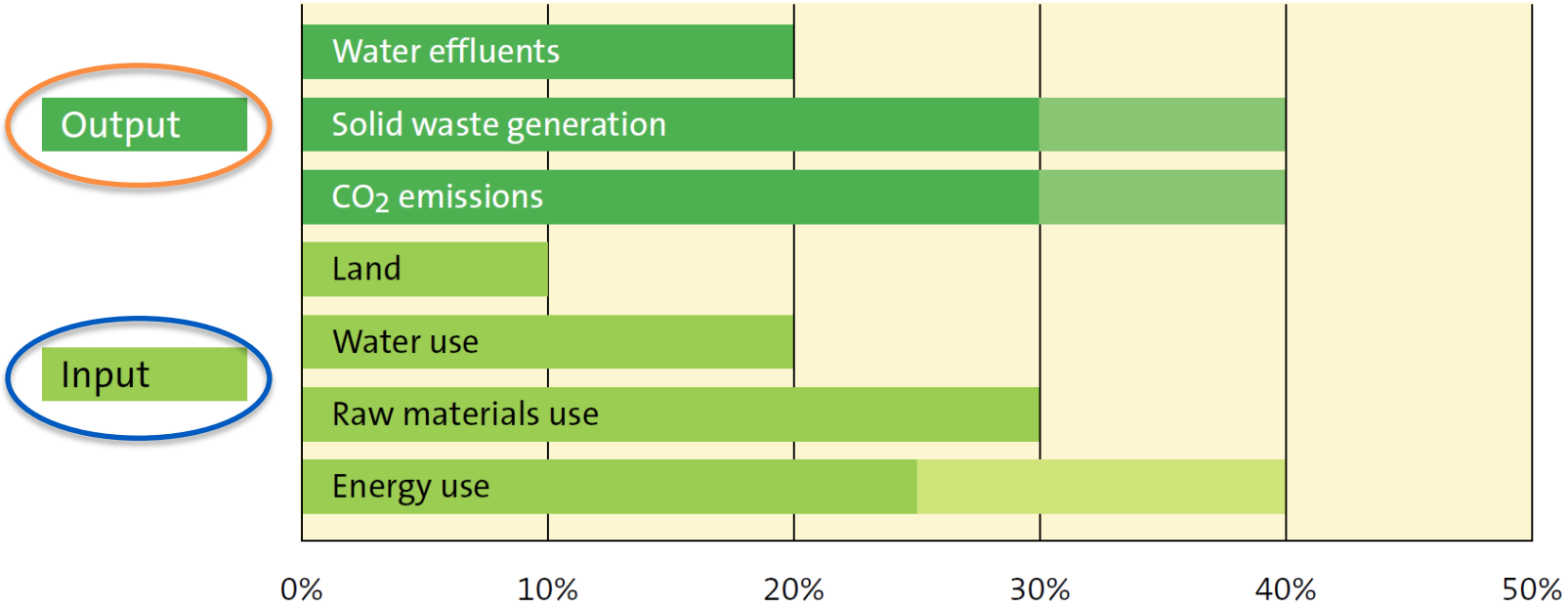

Source: EU Directive on Waste Management and Earth Trends 2007 (Saint-Gobain Isover, 2009)

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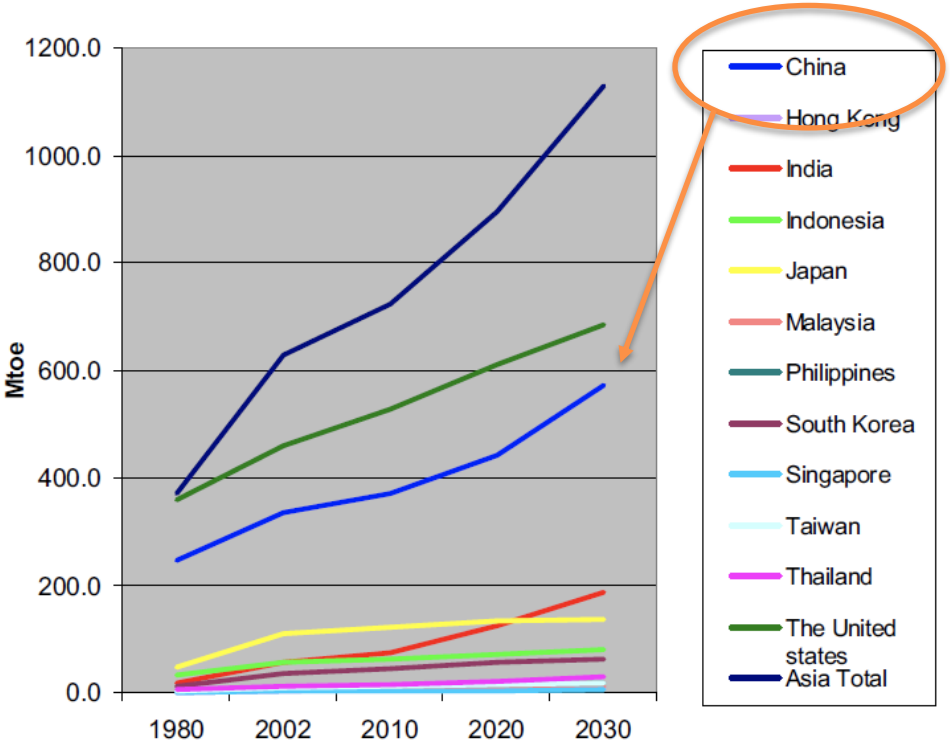
Why is it important?

✓Overview: Impact of the built environment in terms of tonnes of Carbon

✓(Final energy consumption of buildings in Asia)



Source: Trends in Asia's Building Energy Efficiency Policies (Hong & Chiang, 2007)
Asia / World Energy Outlook 2006 (The Institute of Energy Economics Japan, 2006)

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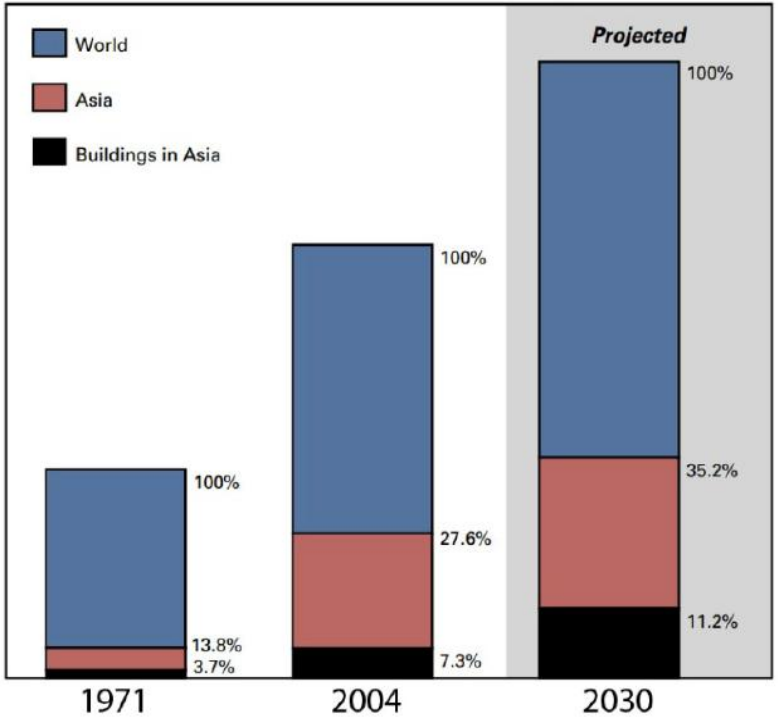
Why is it important?

- ✓Overview: Impact of the built environment
- ✓(Total final energy consumption)

Projections under 'business-as-usual' scenario (to 2030)



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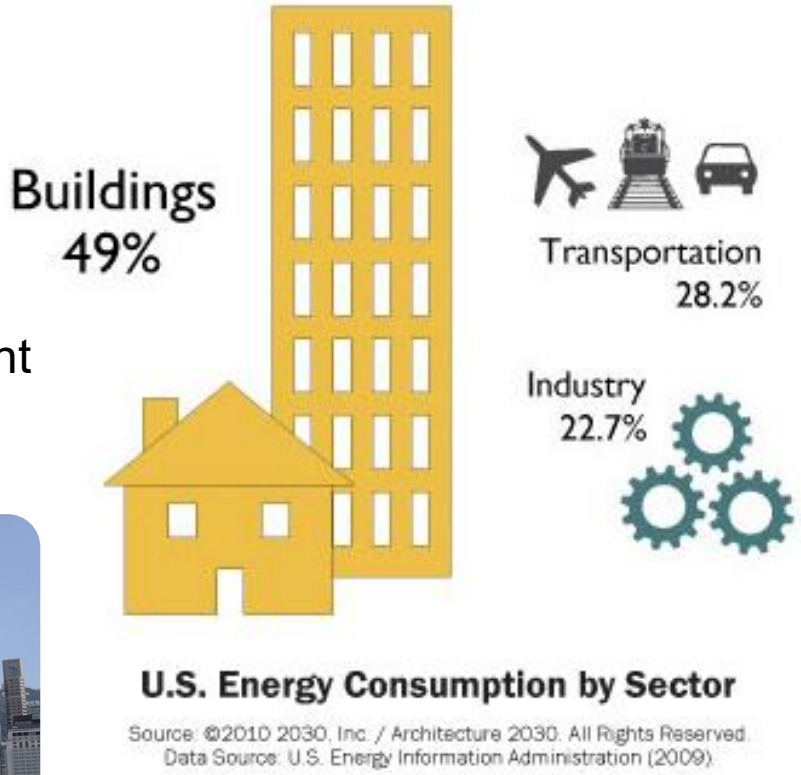
✓Overview: Impact of the built environment

✓(Total final energy consumption)



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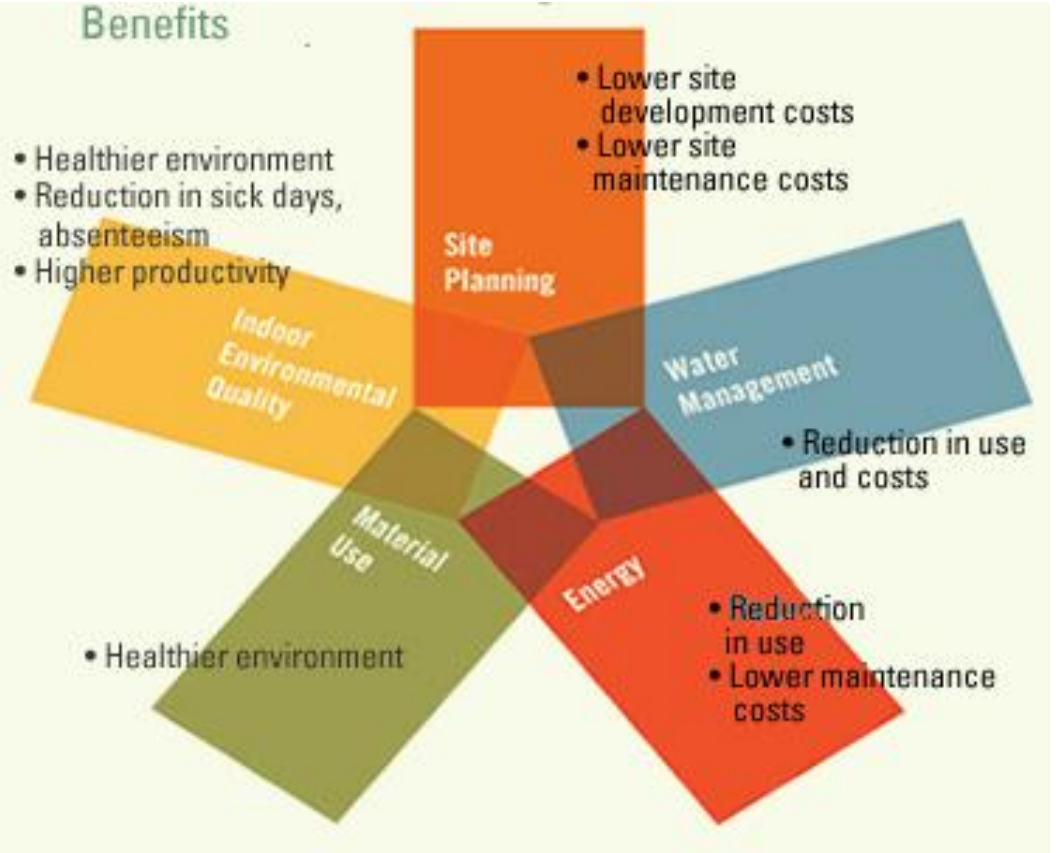
'*Green*' buildings

…it's not just the colour!



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Adapted from:
US Green Building Council (2008)
Chandler Project Management (US)
www.chandlerllc.com/index.html

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Sustainable ‘Green’ buildings

- ✓Worldwide, there is trend towards creating buildings that *use energy carefully* and *without unnecessary waste*

- ✓A building must have both:
  - ✓Minimal impact on the ***natural environment***, and
  - ✓Improve the ***quality of life*** of its occupants.

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Sustainable ‘Green’ buildings

✓Sustainable (or ‘Green’) buildings have:

- ✓Direct benefits – saving operational costs
- ✓Indirect benefits - improving occupants health, comfort, and productivity

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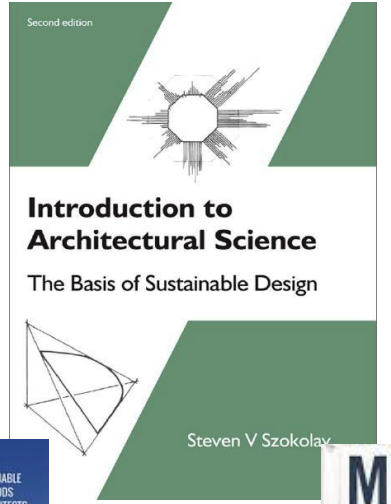
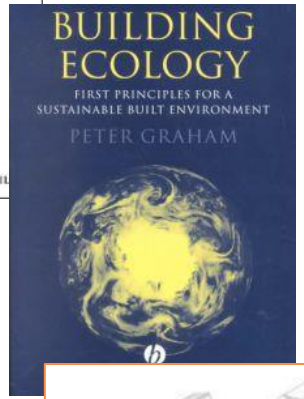
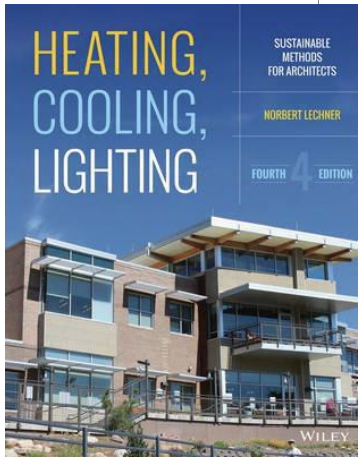
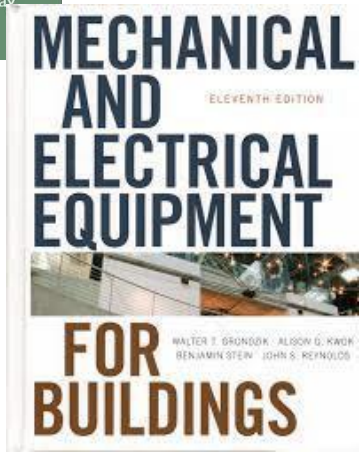
References

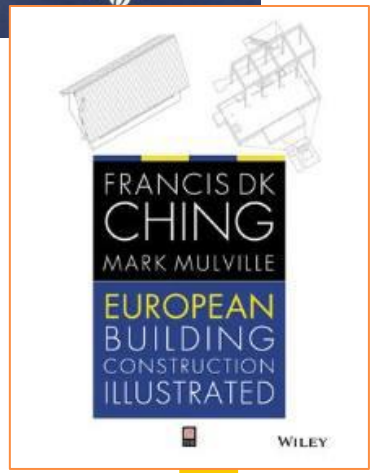
RMIT Library online access











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Thank you

Next time: Climate impacts on Building design

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